BTC TREASURY DESK

# The Bitcoin Whitepaper in 2 Pages

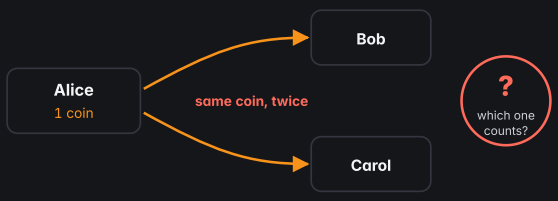
The plain-English gist of Satoshi Nakamoto's 2008 paper, "Bitcoin: A Peer-to-Peer Electronic Cash System."

READ THE ORIGINAL ↗

> **"A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without going through a financial institution."**
>
> Abstract, first sentence
> bitcoin.org/bitcoin.pdf ↗

## 1 The problem: double spending

Digital money is just data, and data can be copied. What stops someone from sending the same coin to two people? Until 2008 the answer was a trusted middleman, like a bank, keeping the official ledger. The paper asks for "an electronic payment system based on cryptographic proof instead of trust."

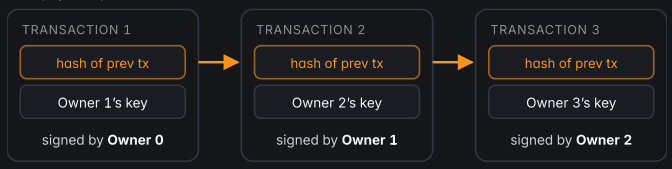



Satoshi's key point: the only way to know a coin wasn't already spent is for everyone to see every transaction, in order.

## 2 Coins are a chain of signatures

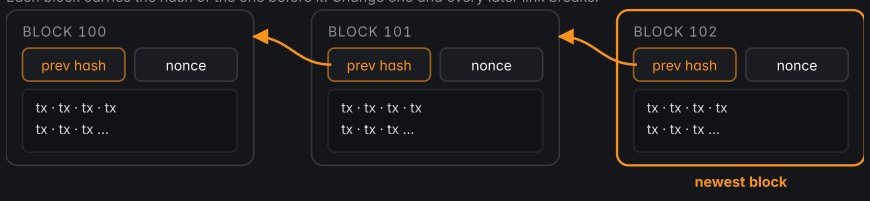
The paper puts it simply: "We define an electronic coin as a chain of digital signatures." To pay you, the current owner signs a message that points to the previous transaction and names your public key. Anyone can check the signatures. Nobody can forge them without the owner's private key.

Each payment points back to the one before and names the next owner.




## 3 A shared clock: blocks, hashes and proof-of-work

Each block carries the hash of the one before it. Change one and every later link breaks.




Transactions are bundled into **blocks**. Each block includes the **hash** (a digital fingerprint) of the block before it, so the blocks form a chain in time order.

To add a block, a computer must find a **nonce** that gives the block a hash below a target. That takes real computing work and electricity to find, but is instant to check. This is **proof-of-work**.

To rewrite history, an attacker would have to redo that work for the changed block and every block after it, and outrun everyone else.

## How the network runs, in the paper's six steps

| 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|
| New transactions are broadcast to all nodes | Each node collects them into a block | Each node works on the proof-of-work | The winner broadcasts its block | Nodes accept it only if every transaction is valid and unspent | Nodes build the next block on top of it |

### The whole paper in one paragraph

Bitcoin replaces the bank's ledger with a public one that everyone keeps a copy of. Coins move by digital signature. Transactions are grouped into blocks, and each block is locked to the one before it by a hash and by real computing work. The honest majority of computing power always extends the longest chain, so rewriting the past gets harder with every block. Miners are paid in new coins and fees for doing this work, which makes playing by the rules the profitable choice.

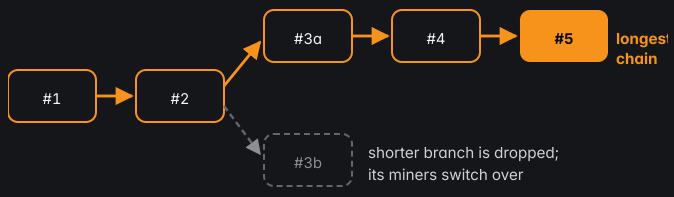
### Key words

**Hash** a fixed-length fingerprint of data · **Node** a computer running the network rules · **Miner** a node doing proof-of-work · **Block** a batch of transactions · **Public / private key** the key others see / the secret that signs

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## 4 The longest-chain rule

Sometimes two miners find a block at almost the same moment, and the chain briefly splits. The rule that settles it: "Nodes always consider the longest chain to be the correct one". Here "longest" means the chain with the most proof-of-work behind it. When the next block lands on one branch, everyone switches to it.




This is why people wait for several confirmations: each new block on top makes an old transaction harder to undo.

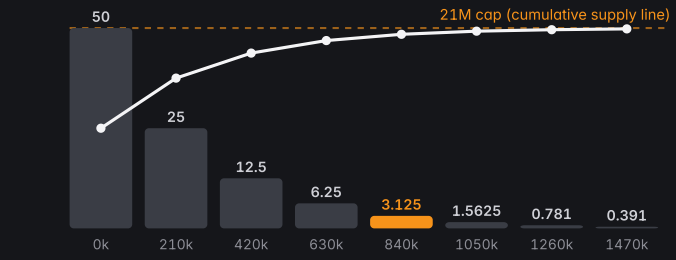
## 5 Incentives: why miners play fair

The first transaction in each block creates new coins for the miner who found it. That reward pays for the work and is how coins first enter circulation. The paper compares it to gold miners spending resources to add gold to circulation. Transaction fees are added on top, and once a set number of coins exists, fees can carry the whole reward.

Voting is by computing power, not by headcount: "Proof-of-work is essentially one-CPU-one-vote." An attacker with huge power should find it more profitable to mine honestly than to undermine the system that pays them.

## 6 The 21 million context

RULES IN CODE, NOT THE PAPER




Bars: new BTC per block, by era start height. Line: total issued.

The whitepaper only says new coins stop after a "predetermined number." The exact schedule lives in Bitcoin Core: **50 BTC** per block to start, **halving every 210,000 blocks**, with a 21 million cap (`MAX_MONEY`). Since block 840,000 the reward is **3.125 BTC**. The next halving is at block **1,050,000**. The chain tip was block 969,929 on Oct 4, 2026 (mempool.space).

## 7 Why it matters for treasury companies

- **Fixed supply, rising holders.** No company can print more bitcoin. Treasury companies compete to own a slice of a capped asset.
- **Scale check.** Strategy's 848,250 BTC (as of Oct 11, 2026) is about **4.0%** of the 21M cap. Strive's 35,595 BTC (8-K filed Oct 14, 2026) is about **0.17%**. (Our math from their 8-Ks.)
- **Trust, but verify.** Anyone can check bitcoin's supply and rules on the chain. Company holdings are self-reported in SEC filings; the chain alone doesn't show who owns which coins.
- **No middleman, no bailout.** The same design that removes the bank also means price swings land fully on the holder: company or you.

## 8 The paper's 12 sections, one line each

OUR SUMMARIES

| | | | | | |
|---|---|---|---|---|---|
| **1 · Introduction** Online payments rely on trusted third parties; that model has real costs. | **2 · Transactions** A coin is a chain of digital signatures. | **3 · Timestamp Server** Hash batches of items to prove they existed, in order. | **4 · Proof-of-Work** Make adding blocks costly and checking them cheap. | **5 · Network** Six steps; nodes follow the longest chain. | **6 · Incentive** New coins and fees reward honest work. |
| **7 · Reclaiming Disk Space** Merkle trees let old data be pruned. | **8 · Simplified Payment Verification** Check payments with block headers only. | **9 · Combining and Splitting Value** One or more inputs; at most two outputs: payment and change. | **10 · Privacy** Public keys, no names. | **11 · Calculations** An attacker's odds fall fast as blocks pile up. | **12 · Conclusion** A system for electronic transactions without relying on trust. |

**Sources.** Satoshi Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System" (2008), bitcoin.org/bitcoin.pdf. All quoted phrases are verbatim from the paper; everything else is our own summary. Supply rules: Bitcoin Core source code, github.com/bitcoin/bitcoin (`GetBlockSubsidy`, `nSubsidyHalvingInterval = 210000`, `MAX_MONEY`), checked Oct 4, 2026. Block height: mempool.space, Oct 4, 2026, 8:48 PM ET. Holdings: Strategy 8-K filed Sep 28, 2026; Strive 8-K filed Sep 28, 2026 (sec.gov).

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